

Adyen Payment File Integration

Automated integration to download and store Adyen payment report files using webhooks and a cloud automation flow (e.g., Power Automate, Logic Apps).

Background

Adyen provides two key report files for payment transaction reconciliation:

Payment Accounting Report

- Available the **next business day** after transactions occur.
- Contains every transaction status update for each PSP reference.
- Useful for early visibility into payment lifecycle events (authorized, captured, refunded, etc.).
- Can be scheduled for automatic generation (daily).

Settlement Details Report

- Generated once Adyen settles transactions with external payment service providers (card schemes, etc.).
- Has **two states**:
 1. **Before batch transfer** — settlement details are available, but no batch number is assigned yet.
 2. **After batch transfer** — Adyen updates the report with the batch number once funds are transferred to the merchant.
- Transactions themselves are not updated between the two states; only the batch number is added.
- Can be scheduled daily, weekly, monthly, or at batch close.

Reference: [Adyen Reporting Documentation \(https://docs.adyen.com/reporting/\)](https://docs.adyen.com/reporting/)

Reports overview

Bookmarked (0) All (9) **Finance (8)** Accounts (1) Performance (0) Point of sale (0) Risk (0) Transactions (0) Others (0)

Settlements

☆ **Aggregate settlement details** Manual
Aggregates all payments that were settled over multiple days. Payments are aggregated by the card schemes and/or payment methods.

☆ **Settlement details** Automated
Transaction details for all settled payments paid out to you by Adyen. Can be scheduled daily, weekly, monthly, and at batch close.

Payments

☆ **Daily finance** Manual
Daily financial summary, with detailed reporting on payments by status, open balances, and payout details.

☆ **Interactive payment accounting** Manual
Overview of payments lifecycle status changes for all transactions. You can select a date range, change time zones, and filter for specific events.

☆ **Interactive payment accounting (Essentials)** Manual
Monthly report on payments lifecycle status changes for all transactions. Only for Essentials users.

☆ **Payment accounting** Automated
Daily report on payments lifecycle status changes for all transactions. Can be scheduled.

☆ **Received payment details** Manual
Daily report on all authorized or refused transactions. Can be scheduled.

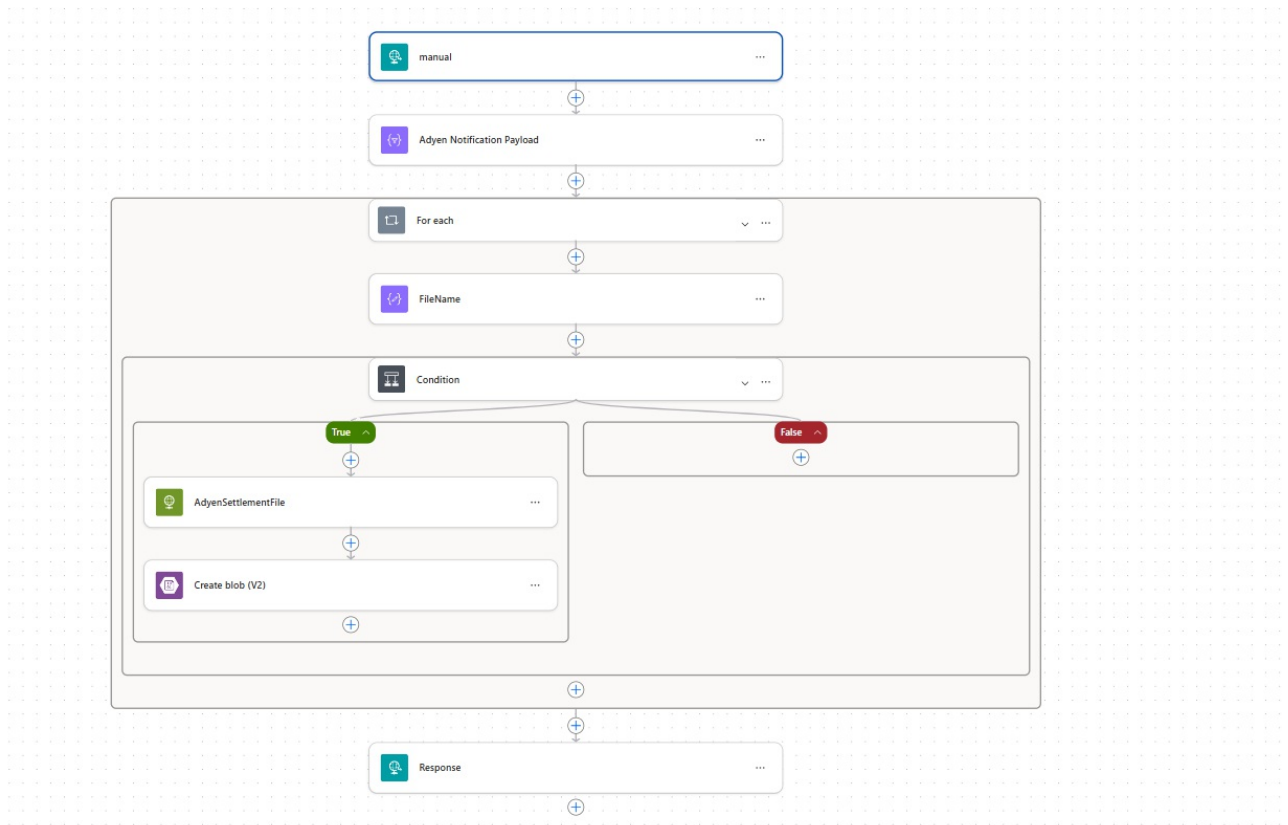
How It Works

Adyen provides a **REPORT_AVAILABLE** webhook notification. When a report is generated, Adyen sends a notification to a configured endpoint. The integration flow then downloads the report file and stores it in Azure Blob Storage.

Architecture Flow

1. Adyen generates a scheduled report (Payment Accounting or Settlement Details).
2. Adyen sends a **REPORT_AVAILABLE** webhook notification to the automation flow's HTTP endpoint.
3. The flow parses the notification payload and iterates over the notification items.
4. For each item, it extracts the file name and determines the report type.
5. Based on a condition (e.g., report type), it downloads the appropriate file from Adyen.

- The file is stored in Azure Blob Storage.
- The flow returns an HTTP 200 response to Adyen to acknowledge receipt.



Setup Guide

1. Set Up Azure Key Vault

The integration stores the Adyen Report Service User API key securely in Azure Key Vault. A PowerShell script (`setup.ps1`) is provided to automate this setup. The secret will be created with a dummy placeholder value initially — you will update it with the real API key in Step 4.

Before running the script, update the following value in `setup.ps1`:

- `$SUBSCRIPTION_ID` — Your Azure Subscription ID. To find it:
 - Go to the [Azure Portal \(https://portal.azure.com/\)](https://portal.azure.com/) → **Subscriptions**, or
 - Run `az account list --output table` in your terminal.

Once updated, run the script:

```
.\setup.ps1
```

The script will:

- Create a Resource Group and Key Vault in Azure.
- Create a placeholder secret in the Key Vault (to be updated with the real API key in Step 4).
- Create an Entra ID (Azure AD) App Registration with **Key Vault Secrets User** role access.
- Output the **Tenant ID**, **Client ID**, and **Client Secret** — save these, as they are needed when configuring connections in Power Automate.

⚠ Important: Copy the **Client Secret** immediately after the script completes — it will not be shown again.

2. Import the Power Platform Solution

This repository includes a Power Platform solution package (`AdyenPaymentFileIntegration_1_0_0_2.zip`) that contains the webhook receiver flow.

- Go to [Power Automate \(https://make.powerautomate.com/\)](https://make.powerautomate.com/) and select your target environment.
- Navigate to **Solutions** in the left-hand menu.

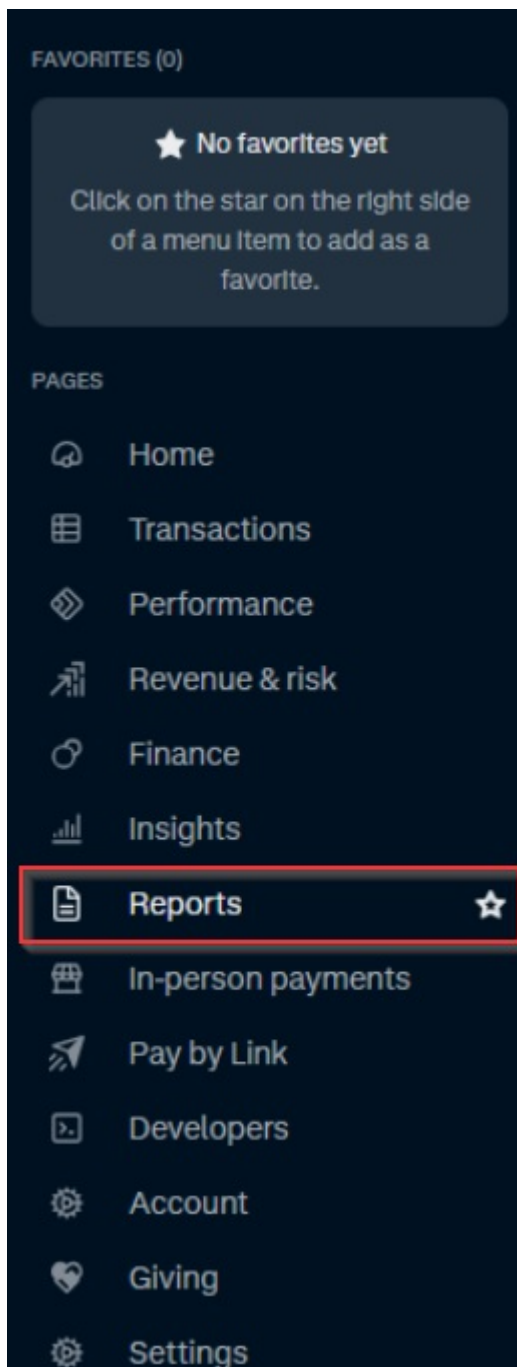
3. Click **Import solution** and upload `AdyenPaymentFileIntegration_1_0_0_2.zip` from this repository.
4. During import, you will be prompted to configure **Connections**. Set up the required connections:
 - **Azure Key Vault** — Use the **Tenant ID**, **Client ID**, and **Client Secret** output from `setup.ps1`.
 - **Azure Blob Storage** — Provide your storage account connection details.
5. Complete the import and verify the flow is enabled.

Once imported, the flow performs the following:

- **Accepts** the incoming POST request from Adyen.
- **Verifies** authenticity using HMAC signature verification.
- **Parses** the `REPORT_AVAILABLE` notification payload.
- **Downloads** the report file from Adyen using the Report Service User credentials (retrieved from Key Vault).
- **Stores** the file in Azure Blob Storage.
- **Returns** HTTP 200 promptly to Adyen (long-running operations should not block the response).

3. Configure Report Generation in Adyen

1. Log in to the [Adyen Customer Area \(https://ca-live.adyen.com/\)](https://ca-live.adyen.com/).
2. Navigate to **Reports** in the left-hand menu.



3. Locate the reports you need (typically **Payment Accounting** and/or **Settlement Details** under the Finance tab).
4. Configure them to be generated **automatically** (daily/weekly) by clicking **Generate** and selecting a schedule.

4. Create Report Service User Credentials

To download reports programmatically, you need dedicated API credentials:

1. In the Customer Area, go to **Developers** → **API Credentials**.
2. Create a new credential set for a **Report Service User**.
3. Assign the appropriate report download roles/permissions.
4. Copy the generated **API key**.

Reference: [Adyen API Credentials \(https://docs.adyen.com/development-resources/api-credentials/\)](https://docs.adyen.com/development-resources/api-credentials/)

Update the Key Vault Secret

Now update the placeholder secret created in Step 1 with the real API key:

```
az keyvault secret set --vault-name PaymentReconkeyvault --name ReportUserAPIKey --value "<your-adyen-report-user-api-key>"
```

Replace <your-adyen-report-user-api-key> with the API key you just copied from the Adyen Customer Area.

⚠ Never commit credentials to source control.

5. Configure the REPORT_AVAILABLE Webhook

1. In the Customer Area, go to **Developers** → **Webhooks**.
2. Create a new webhook and subscribe to the **REPORT_AVAILABLE** event type.

Webhook details

General

Server configuration

Security

Events

REPORT_AVAILABLE

3. Set the endpoint URL to your automation flow's HTTP trigger URL.
4. Enable **HMAC signature verification** for security.

Reference: [Adyen Webhooks \(https://docs.adyen.com/development-resources/webhooks/\)](https://docs.adyen.com/development-resources/webhooks/)

Webhook Payload Example

```
{
  "live": "true",
  "notificationItems": [
    {
      "NotificationRequestItem": {
        "eventCode": "REPORT_AVAILABLE",
        "pspReference": "report_file_name.csv",
        "reason": "https://ca-live.adyen.com/reports/download/..."
      }
    }
  ]
}
```

Security Considerations

- **HMAC Verification:** Always verify the HMAC signature on incoming webhook requests to ensure they originate from Adyen.
- **Credential Storage:** Store API credentials and HMAC keys in a secure vault (e.g., Azure Key Vault). Never hardcode secrets.
- **HTTPS Only:** Ensure the webhook endpoint uses HTTPS.
- **Acknowledge Quickly:** Return HTTP 200 to Adyen within a few seconds. Offload heavy processing (file download, storage) to asynchronous steps.

References

- [Adyen Reporting Overview \(https://docs.adyen.com/reporting/\)](https://docs.adyen.com/reporting/)
- [Adyen Webhooks \(https://docs.adyen.com/development-resources/webhooks/\)](https://docs.adyen.com/development-resources/webhooks/)
- [Adyen API Credentials \(https://docs.adyen.com/development-resources/api-credentials/\)](https://docs.adyen.com/development-resources/api-credentials/)
- [Power Automate Documentation \(https://learn.microsoft.com/en-us/power-automate/\)](https://learn.microsoft.com/en-us/power-automate/)